

Hyperparameter Tuning and Model Evaluation

Beyond Accuracy: A Clinical Validation Approach to Cardiovascular Disease Prediction Using Machine Learning

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DSE6311 Capstone

This project evaluates machine learning model performance through a clinical validation lens rather than standard accuracy metrics. To assess clinically appropriate risk, how should model performance be evaluated, given the trade-off between sensitivity and specificity?

Model optimization and validation will balance the delicate trade-off between false negatives and false positives. The final model may not be the most accurate based solely on accuracy/ROC-AUC metrics.

The initial hypothesis predicted that tuned ensemble methods would outperform Logistic Regression on sensitivity and ROC-AUC. This hypothesis was not supported by held-out test set results. Logistic Regression with Cost-Sensitive Learning consistently performed better than both tuned Random Forest and XGBoost on the held-out test set across all clinically relevant metrics. This finding is consistent with known limitations of high-capacity models on small datasets and supports selecting LR CSL as the final model (see Table 4 for all model comparison metrics). The optimal model will be defined as the sensitivity-specificity combination that best aligns with clinical risk thresholds and patient outcomes.

Methods:

- Random Forest and XGBoost models evaluated at baseline without hyperparameter tuning.
- RF and XGBoost baseline models compared with and without CSL weighting.
- RF Standard Baseline, default 100 number of trees
- Models to use Grid Search and StratifiedRepeatedKfold(k=10, repeats=2) for hyperparameter tuning.
- Scoring metric for grid search optimization set to recall – consistent with clinical sensitivity target
- Parameters RF tuned: n_estimators, max_depth, min_samples_leaf, class_weight
- Parameters XGB tuned: learning_rate, max_depth, n_estimators, scale_pos_weight, subsample
- Parameters XGB second round tuning: min_child_weight, gamma, reg_alpha, reg_lambda plus previous round
- SVM linear and RBF were added as comparison models to assess metric for small dataset size. Baseline and GridSearchCV methods applied.

Results:

RF Standard Baseline - Training

- Default RandomForest showed perfect training scores (1.000 across all metrics) in Cv, indicating severe overfitting.
- Test recall of 0.734, below clinical target and below LR baseline.
- Train-test gaps exceed 10% across all metrics – hyperparameter tuning required.
- High variance problems are inherent in unconstrained decision trees.

RF Standard Baseline – Test Evaluation

- Underperformed LR CSL on all baseline metrics.
- Sensitivity 71.8% with 11 missed CVD cases vs 8 for LR CSL.
- ROC – AUC 0.0909 vs 0.933 LR CSL.
- Consistent with overfitting identified in cross-validation.
- Results motivate hyperparameter tuning to constrain tree growth and reduce variance.

RF CSL Baseline - Training

- CSL performs slightly worse than the RF baseline model, both show severe overfitting with perfect training scores across all metrics.
- Hyperparameter tuning is required before comparing RF variants.

RF CSL Baseline – Test Evaluation

- RF CSL Baseline performed worse than RF Standard on all metrics before tuning – sensitivity dropped from 71.8% to 66.7% with 13 missed CVD cases.
- This counterintuitive finding is attributed to the interaction between severe overfitting and class weighting on unconstrained trees.
- Results confirm hyperparameter tuning must precede any meaningful comparison between standard and CSL variants for tree- based models.

RF Tuned – Training

- Hyperparameter tuning successfully resolved severe overfitting.
- Best params: {'class_weight': 'balanced', 'max_depth': 3, 'min_samples_leaf': 4, 'n_estimators': 100} Best recall: 0.766
- Train – test gaps reduced to under 8% across all metrics.
- Recall improved from 0.724 to 0.756.
- ROC – AUC of 0.900
- Sensitivity still below target of $\geq 85\%$.

RF Tuned – Test Evaluation

- Accuracy 82.1%, Recall 74.4%, Specificity 88.9%, ROC -AUC 0.928 (LR CSL 0.933)
- 10 missed CVD cases, same as LR standard, worse than LR CSL.
- RF Tuned does not outperform LR CSL baseline on any metric.
- Performance gap possible due to small dataset size, RF benefits from larger datasets.

XGB Standard Baseline - Training

- Severe overfitting as seen with untuned RF model.
- Hyperparameter tuning required.
- Small dataset poorly suited for this type of model

XGB Standard Baseline – Test Evaluation

- Provides neat same results as the RF Tuned model, but underperformed the LR CSL on all baseline metrics.
- Sensitivity 74.4% with 10 missed CVD cases vs 8 for LR CSL.
- ROC – AUC 0.882 vs 0.928 RF Tuned. Classifies patients the same but RF Tuned has better calibration.

XGB CSL Baseline - Training

- CSL performs slightly worse than the XGB baseline model, both show severe overfitting with perfect training scores across all metrics. Same as with RF modeling.

XGB CSL Baseline – Test Evaluation

- XGB CSL Baseline performed worse than XGB Standard on all metrics before tuning – sensitivity dropped from 74.4% to 66.7% with 13 missed CVD cases.
- This counterintuitive finding is attributed to the interaction between severe overfitting and class weighting, same as seen in RF.
- Results confirm hyperparameter tuning must precede any meaningful comparison between standard and CSL variants for tree- based models.

XGB Tuned – Training

- Best parameters: {'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 100, 'scale_pos_weight': 1.1734693877551021, 'subsample': 0.8} – still heavy overfitting
- Second round of tuning: Best params: {'gamma': 0.3, 'min_child_weight': 5, 'reg_alpha': 0.1, 'reg_lambda': 2} Best recall: 0.817
- Second round of tuning significantly improved metrics, train test gap 0.054, recall 0.817, ROC – AUC 0.907
- XGBoost tuning recall progression: 0.747 → 0.776 → 0.817
- Now best performing model in training

XGB Fully Tuned – Test Evaluation

- Two rounds of hyperparameter tuning required to address severe overfitting.
- Final test set recall was 74.4%, similar to RF Tuned.
- Discrepancy between CV recall (0.817) and test set recall (0.744) attributed to small test set size (n=39 CVD patients) where individual patient classifications can affect aggregate metrics.
- ROC – AUC 0.917 shows strong discriminative ability.
- 10 missed CVD cases, same as RF Tuned, worse than LR CSL.
- Despite superior CV performance, XGBoost did not outperform LR CSL on held-out test set.

Overall findings so far are consistent with the literature that demonstrates logistic regression performs comparably or better than machine learning approaches for predicting major chronic diseases (Nusinovici et al, 2020). This project can confirm that model complexity does not necessarily translate into improved clinical prediction performance. A finding from Couronne et al 2018, shows that RF generally produces better results, but on diverse datasets.

SVM – Linear - Standard

The training data in cross-validation showed some minor overfitting, which will be monitored, but GridSearch hyperparameter tuning will be applied.

The test evaluation at baseline performed well in cross-validation, exceeding the AUC of LR CSL at 0.946, and specificity exceeded LR CSL at 0.933 vs 0.911. With that, we see the sensitivity trade-off: 0.769 (SVM) vs 0.795 (LR CSL), leading to 9 false negatives vs 8 in the LR CSL.

SVM – Linear – Tuned

After tuning and applying hyperparameters: Best C: {'C': 1} Best CV recall: 0.761, SVM improved recall from 0.769 to 0.795, recovering one additional missed CVD case. This comes from the penalty to penalize false negatives more heavily. Again, the tradeoff of specificity was seen with a drop to 0.911. AUC remained stable at 0.944, and this model produced identical recall and specificity to LR CSL.

SVM – RBF - Standard

This model, when trained and tested, performed similarly to the SVM linear model, but with lower AUC and larger cross-validation overfitting. RBF tuning was omitted since it showed no performance benefit with added complexity. This decision was made on the principle that the simplest model with equal results is preferred over a complex one.

ROC – AUC Curve Analysis

ROC – AUC analysis shows that LR CSL and SVM linear CSL are both capable of achieving the clinical sensitivity target of $\geq 85\%$. Both models cross the threshold at false positive rates of ~ 0.20 . Threshold optimization is identified as the next key step for meeting clinical performance targets at the default threshold of 0.05 (see Figure 17).

SHAP Feature Importance

SHAP analysis of LR CSL model identified `thal_7.0`, `ca`, and `cp_4` as strongest predictors of CVD presence. The engineered `st_severity` ranked 5th. Features at the bottom of importance include `restecg_1`, `thal_6.0`, and `cp_2`, and could possibly warrant removal in future iterations. (see Figure 18)

Threshold Analysis

At default of 0.50, no model achieved clinical targets. Threshold analysis revealed all four tuned models can meet both targets with optimization. LR CSL achieved the highest sensitivity of

0.949 at threshold 0.25, missing only 2 CVD cases, while RF Tuned showed the best sensitivity/specificity balance at threshold 0.40, 0.897, and 0.844, respectively. (see Figure 19)

SVM Linear CSL showed a marginally higher AUC (0.944 vs 0.933), threshold analysis revealed that LR CSL achieved better sensitivity at an optimized threshold of 0.25, missing only 2 CVD cases, compared to 3 in SVM. In a clinical setting where minimizing misses is the primary objective, the reduction in false negatives outweighs the difference in AUC.

Decision Curve Analysis

DCA confirmed clinical use in all four models across threshold range of 0.10 – 0.40. All model curves above the treat-all and treat-none baselines. LR CSL and SVM Linear CSL show highest net benefit, with LR CSL as the recommended model for clinical deployment. These result show that using the model to guide decision making would be better than treating all or treating no patients. (see Figure 20)

Discussion and Next Steps

Analysis showed that a logistic regression model with cost-sensitive learning, evaluated through the lens of clinical context, can meet performance targets for CVD classification. The hypothesis for this project was that ensemble methods would outperform LR, but the findings showed they did not. The core elements of the hypothesis were validated: the “best model” is the one that best serves the patient, which requires evaluating sensitivity, specificity, false-negative cost, threshold optimization, and clinical use. A model with lower AUC can be the superior clinical choice.

Evaluated at an optimized threshold of 0.25, LR CSL achieved a sensitivity of 0.949, while maintaining specificity above the clinical minimum. DCA confirmed clinical usefulness, providing evidence that model deployment would produce better patient outcomes than treating all or none of the patients. Primary limitations remain the small sample size, the age of the data, and the environment of data collection (exercise stress testing domain, not general population). Future work should focus on using a larger and current dataset, and potentially incorporating modern updates in CVD detection testing. No further modeling is planned for this project; the analysis is complete, and the findings are sufficient to address the research question.

Appendix

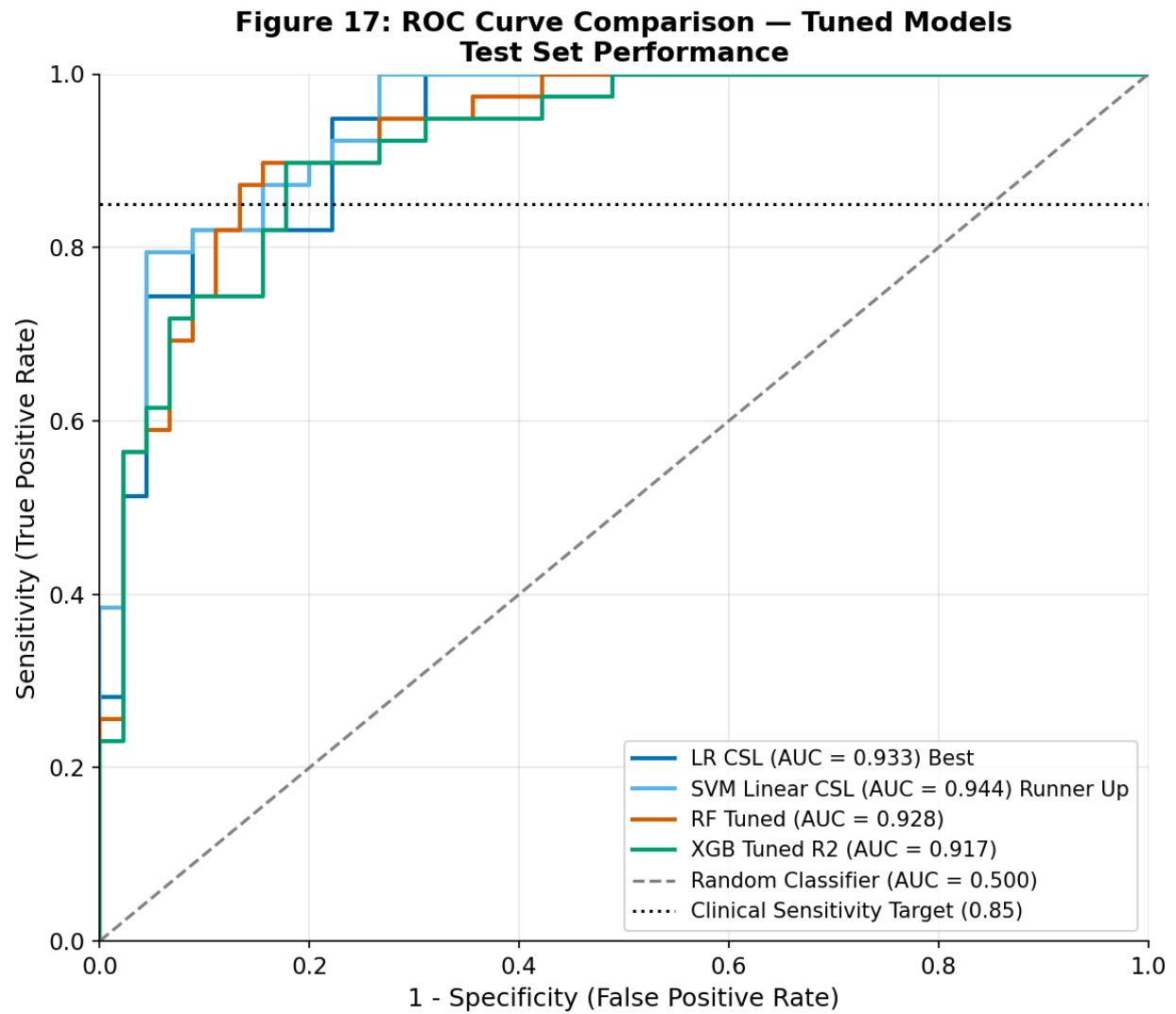


Figure 17: ROC – AUC of final tuned models. LR CSL and SVM Linear CSL best and runner – up models.

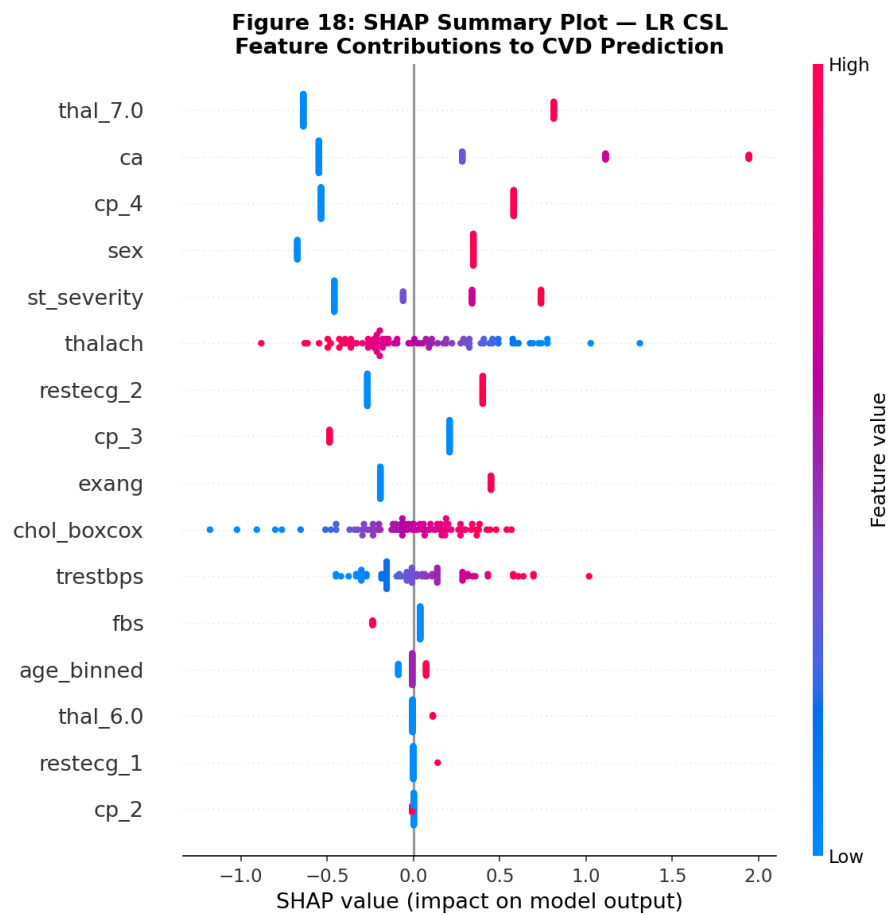


Figure 18: SHAP Feature Importance – LR CSL Model. Blue for categorical = 0 indicator

**Figure 19: Threshold Analysis — Sensitivity & Specificity Across Thresholds
Tuned Models — Test Set**

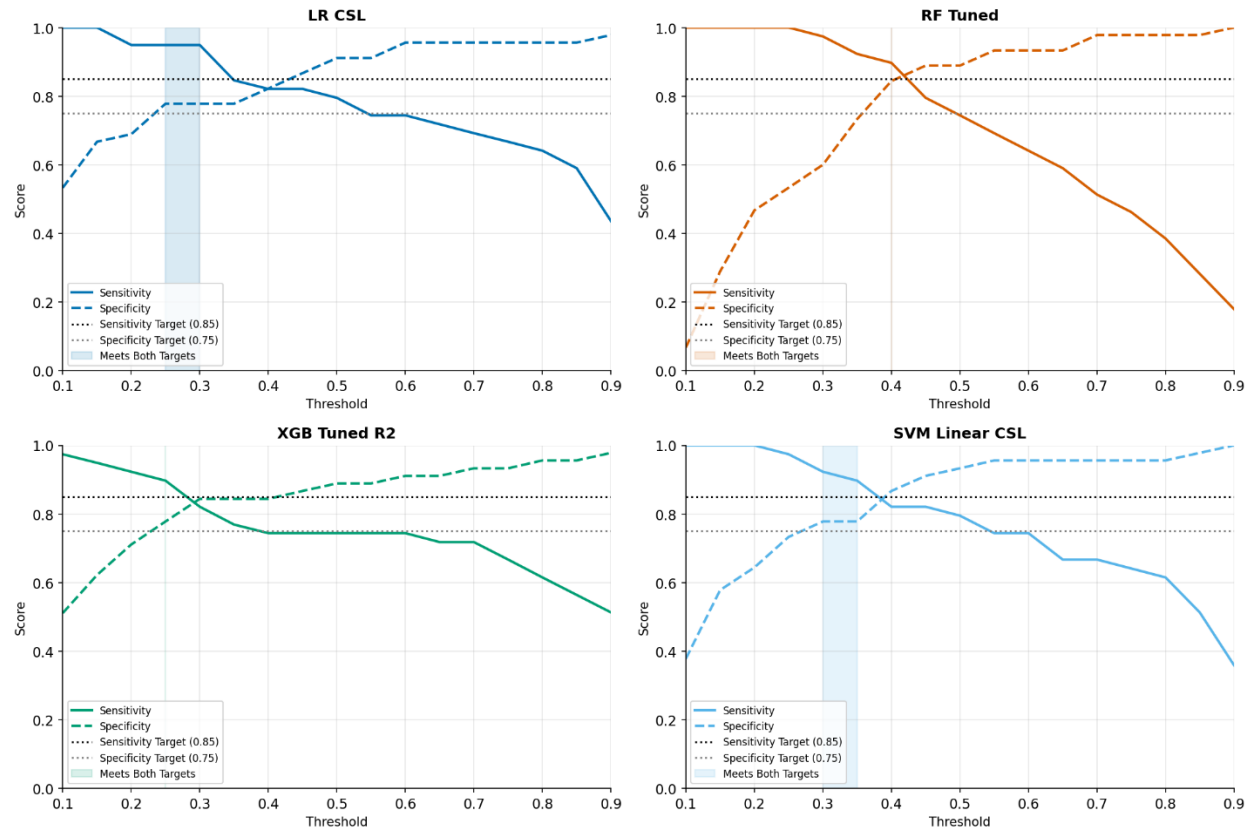


Figure 19: Threshold Analysis of the four tuned Models

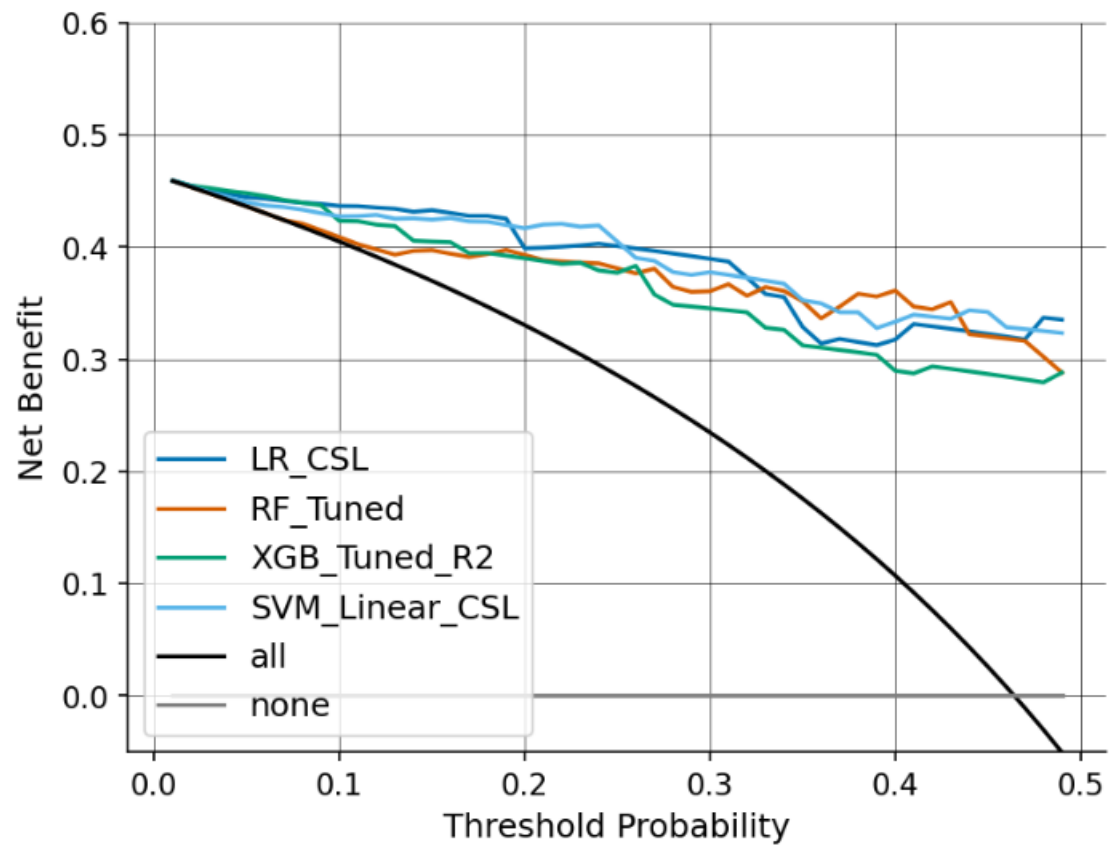


Figure 20: Decision Curve Analysis results

Table 4: Full Model Comparison - Baselines vs Tuned Models - Test Set Results

	Accuracy	Recall	Specificity	F1	AUC	FN	FP
LR Standard	0.833	0.744	0.911	0.806	0.932	10	4
RF Standard	0.810	0.718	0.889	0.778	0.909	11	5
XGB Standard	0.821	0.744	0.889	0.795	0.882	10	5
SVM Linear	0.857	0.769	0.933	0.833	0.946	9	3
SVM RBF	0.857	0.769	0.933	0.833	0.940	9	3
LR CSL	0.857	0.795	0.911	0.838	0.933	8	4
RF Tuned	0.821	0.744	0.889	0.795	0.928	10	5
XGB Tuned R2	0.821	0.744	0.889	0.795	0.917	10	5
SVM Linear CSL	0.857	0.795	0.911	0.838	0.944	8	4

Table 4: Showing all metrics for all models. Highlighted models are best performers.

Data Dictionary

Variable	Type	Description	Values/Range
age_binned	Categorical	Age group by categories to address BT violation	0=Young (<45), 1=Middle (45-60), 2=Older (>60)
sex	Categorical	Biological sex	0 = Female, 1 = Male
cp	Categorical	Chest pain type	1 = Typical Angina, 2 = Atypical Angina, 3 = Non-Anginal Pain, 4 = Asymptomatic
trestbps	Continuous	Resting blood pressure on admission (mmHg)	94 - 200, note : standardized via StandardScaler during preprocessing
chol_boxcox	Continuous	Box - Cox transformed cholesterol serum	standardized Original range: 126-564 mg/dl, transformed and standardized via StandardScaler
fbs	Categorical	Fasting blood sugar > 120 mg/dl	0 = Normal, 1 = Elevated
restecg	Categorical	Resting electrocardiographic results	0 = Normal, 1 = ST-T Abnormality, 2 = LV Hypertrophy
thalach	Continuous	Maximum heart rate achieved during exercise (bpm)	71 - 202, note: standardized via StandardScaler during preprocessing
exang	Categorical	Exercise induced angina	0 = No, 1 = Yes
st_severity	Categorical	ST depression and slope of peak exercise severity index	0 = Normal, 1 = Equivocal, 2 = Positive, 3 = Strongly Positive
ca	Categorical	Number of major vessels colored by fluoroscopy	0 - 3
thal_	Categorical	Thalassemia type	_3.0 = refrence category, _6.0 = Fixed Defect, _7.0 = Reversible Defect
target	Binary Target	Cardiovascular disease diagnosis	0 = No CVD, 1 = CVD Present

References

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Logistic regression was as good as machine learning for predicting major chronic diseases
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